JellyComet

JellyPitchBtw manual

Pitch correction for one voice, steered by a jellyfish. One puck sets how fast a note is pulled in and how far off its note a voice may be and be left alone, and a display shows the voice as sung and as corrected.

Version 1.3.0. To install it, see the README.

Contents
Getting started
The controls
How to use it
Good to know
Troubleshooting

Contents

- Getting started
- The controls
- How to use it
- Good to know
- Troubleshooting

Getting started

Put JellyPitchBtw on a track with one voice on it: a lead vocal, a backing vocal, a solo instrument that plays one note at a time. Choose the song's key and scale at the top left, play the track, and watch the display: the blue line is the voice as sung, the bright line as it comes out, and each note it is pulled to sits behind them as a bar.

At the defaults (the puck at the bottom of the pad, half way across) a note is pulled in over 25 ms and every note is corrected. Drag the puck to the bottom left corner to hear the hard, robotic snap; up and to the right for a correction you barely notice.



The controls

From the top left of the window.

Control	Range	Default	What it does
Power switch (top right)	On / Off	On	Off, the voice passes through uncorrected, as late as the latency the plugin reports, so the track stays in time.
Key and scale	C to B; 13 scales	C, Chromatic	The notes a voice is pulled to. Chromatic is every note: plain tuning.
VOICE (beside the scale)	Low, Normal, High	Normal	Where the voice is looked for: Low 50 to 400 Hz, Normal 80 to 800 Hz, High 150 to 1500 Hz.
The pad: SPEED (left to right)	0 to 200 ms	25 ms	How fast a new note is pulled in. 0 ms snaps at once; 200 ms glides.
The pad: HUMANIZE (bottom to top)	0 to 100 %	0 %	How far off its note a voice may be and be left alone, up to 50 cents at 100 %. 0 % corrects every note.
STRENGTH (display)	0 to 100 %	100 %	How much of the correction is applied. 0 % leaves the pitch as sung.
DETUNE (display)	−100 to +100 cents	0 ct	Moves the corrected voice off the note, for a slightly flat or sharp double.
A4 (display)	415 to 466 Hz	440.0 Hz	The tuning reference, for a band or a recording tuned off 440.
Zoom (top right)	100 to 200 %	100 %	The window's size, kept with the project.

How to use it

The pad

The puck's two directions are the two things that make correction sound natural or not.

- **Left and right is SPEED:** how long a new note takes to be pulled in. Fast (left) is the stepping, robotic effect; slow (right) lets the singer's own slide into a note through.
- **Up and down is HUMANIZE:** how far off its note a voice may be and be left alone. At the bottom every note is corrected; higher up, notes already close stay as sung, so a held note keeps its life and only the ones that are clearly off get moved.

Press anywhere on the pad and the puck jumps there and follows you; press on the puck itself to take hold of it where it is. Shift makes a drag ten times finer, and a double-click puts the puck back to the defaults. The values under the pad move on their own: drag one up or down, use the wheel over it, right-click it to type a value, or double-click to reset it.

The note follows the pitch as it is sung: a wide vibrato that reaches past the half-way point swings to the next note and back, the sound of a hard-tuned vocal. Settings to start from:

For	SPEED	HUMANIZE
The hard, stepping effect	0 to 1 ms	0 %
A tight pop vocal	10 to 25 ms	0 to 20 %

For	SPEED	HUMANIZE
A natural fix you shouldn't hear	50 to 120 ms	20 to 40 %

VOICE

VOICE, beside the key and scale, says where to look for the voice. Click it for its list: Low, Normal and High, each with its range in Hz. Normal fits most singers; take Low for a bass voice that the plugin misses, and High for a high soprano or a whistle. The latency is the same in all three. VOICE isn't automatable: set it and leave it.

The display

The display shows the last four seconds, the newest at the right: the blue line is the voice as sung, the bright line as corrected, the glow between them the correction, and the bars behind them the notes it was pulled to. A rail lies on each note of the scale, the key's brighter. At the lines' head a chip names the note and how far the voice is off it, `G1 +7 ct`.

The values along its top and bottom edges are its controls: drag one up or down, use the wheel over it, or double-click it to reset; Shift is finer.

- STRENGTH below 100 % pulls only part of the way, for a voice that is helped rather than fixed.
- DETUNE moves the result off the note: a copy of the vocal at +8 ct under the original thickens it.
- A4 is the tuning reference.

The jellyfish

The jellyfish listens too. Its bell pulses and lights with the level, and its tentacles hang calm while the voice is on its note (within 5 cents) and stir more the further it is off, all the way from 50 cents. Switched off, it rests.

Good to know

- Latency: 0.25 ms in every VOICE. The host is told, and lines the track up. While a voice is being tuned the sound runs a little further behind, 3.8 ms on average: to move a note the plugin repeats or drops one cycle of the voice now and then, so it is about 2 to 3 ms for a high voice and 5 to 6 ms for a low one.
- Automation: every parameter but VOICE can be automated.
- From 1.2: every setting is kept. The latency was 1 ms.
- From 1.1: every setting is kept. The latency was 3.75 to 10.75 ms by VOICE.
- From 1.0: a project saved with 1.0 opens with its key, scale, SPEED, STRENGTH, DETUNE, A4 and VOICE. REALTIME, VIBRATO, FORMANT and HOLD are gone, and HUMANIZE starts at 0 %.
- Saved with the project: every setting and the zoom.
- CPU: 0.13 % of a core at the defaults. It needs a CPU with AVX2 (Intel Core from 2013, AMD from 2015).
- One voice: chords, a choir on one track or a mix go wrong. Correct each voice on its own track.
- Zoom: the project remembers it, and a new window opens at the zoom you last chose in any JellyComet plugin.
- Hosts: the CLAP passes `clap-validator`.
- Display: X11, or XWayland on a Wayland desktop, with a Vulkan driver.

Troubleshooting

- **No correction, the display stays empty:** check that the power switch is on and that the voice is in VOICE's range; a low voice may need Low.
- **Notes jump to the wrong note:** the key or scale is wrong for the song. Chromatic corrects to the nearest note whatever the key.
- **The track sounds late against the others:** the host has to compensate the plugin's latency; check that its latency compensation is on.
- **The plugin doesn't load:** it needs a CPU with AVX2 (Intel Core from 2013, AMD from 2015) and can't run on an older one; `grep -c avx2 /proc/cpuinfo` prints 0 there. 1.1.0 runs without it.
- **The window stays black:** the editor needs a Vulkan driver (`vulkaninfo` should list your GPU).